AKENTEN APPIAH-MENKA UNIVERSITY
of Skills Training *and* Entrepreneurial Development

ICT233;

Web Technologies and Design

Lecture Two

1. **Web Browsers and Servers**
2. **Web Browser Reference Architecture.**
3. **The Web Hosting Process**
4. **The Web Design Process ***

Today's Lecture

- **Web Browsers and Web Severs**
- **Web Browsers Reference Architecture**
- **The Web Hosting Process**
- **The Web Design Process**

Lecture Objectives

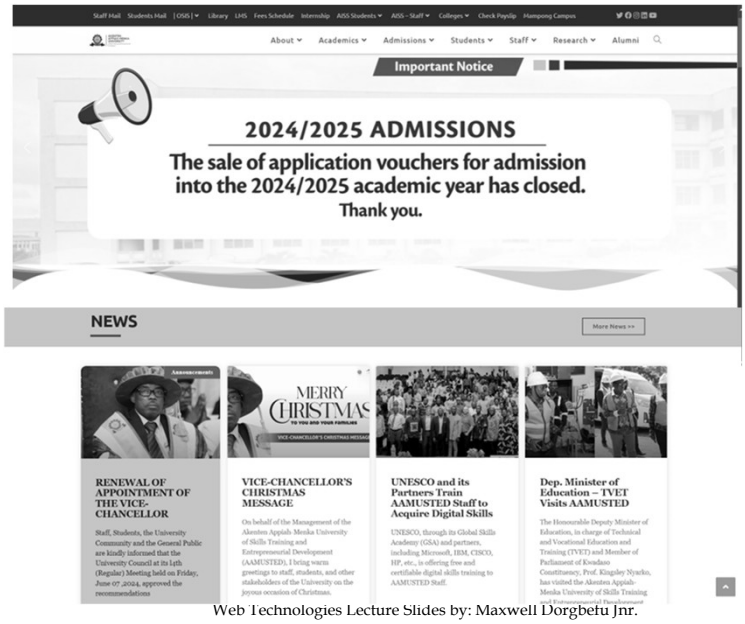
- By the end of the lecture, the student will be able to:
  - Describe web browsers and their functions.
  - Describe the components of web browser reference architecture.
  - Describe the key functions of web servers.
  - Explain the various stages of the web design process.
  - Describe the web hosting processes.

Web Browsers - 1

- A **web browser** is an application software for accessing, traversing, retrieving, and rendering information resources on the world wide web.
- A web browser can also be defined as an application software or program designed to enable users to access, retrieve and view documents and other resources on the web.

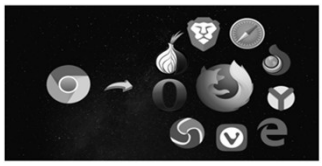
What Web Browsers do -- 1

What Web Browsers do -- 2



Web Browsers - 2

- Although browsers are primarily intended to use the World Wide Web, they can also be used to access information provided by web servers in private networks or files in file systems.
- The major web browsers are: ***Chrome, Firefox, , MS Edge, Opera, and Safari.***
- ***Find list of browsers and their logos at*** ***https://www.webdevelopersnotes.com/browsers-list***



Reference Architecture for Web Browsers

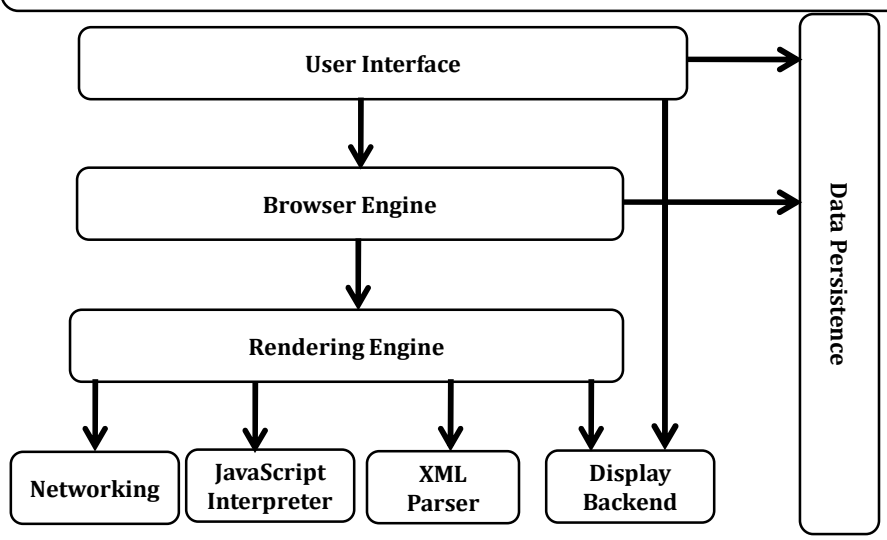


Fig 2.1: Reference Architecture for Web Browsers

Browser Architecture - 1

- The web browser architecture consist of eight major subsystems and the relationships that exist between them:

1. **The User Interface** lies between the user and the browser engine.

- This includes the address bar, back/forward buttons, bookmarking menu etc.
- Basically every part of the browser display except the main window where you see the requested page.

Browser Architecture - 2

2. **The Browser Engine** subsystem provides an interface for the rendering engine.

- It is responsible for loading a given URI and supports browsing actions such as forward, back, refresh/reload.
- It makes querying and manipulation of rendering engine settings possible.

3. **The Rendering Engine** subsystem is responsible for the visual representation of URI inputs.

- It displays HTML and XML documents, including embedded contents (e.g. images, videos) if there are any.
- It is responsible for positioning of page elements within the page and sets up the page layout. It includes the HTML parser.

Browser Architecture - 3

4. **Networking** subsystem is where file transfer protocols FTP and HTTP are enforced.
- It resolves **MIME** types of files.

5. **JavaScript Interpreter** assesses Java scripts which may be embedded in web pages.
- Some JavaScript functionalities include pop up windows and these may be blocked/disabled by the browser engine or rendering engine for security purposes.

Browser Architecture - 4

6. **The XML Parser** subsystem is one of the most reusable subsystems in the architecture.
- Almost all browser implementations enforce an existing XML Parser rather than building their own from scratch.

7. **The Display Backend** subsystem offers drawing and windowing primitives, a set of user interface widgets, and a set of fonts. May be tied closely to the OS.

8. **The Data Persistence** subsystem stores various data associated with the browsing session on disk.
- This may be high-level data such as bookmarks or toolbar settings, or it may be low-level data such as cookies, security certificates, or cache.

Web Servers - 1

- A **Web server** can refer to either the hardware (the computer) or the software (the computer application) that helps to deliver web contents that can be accessed through the Internet.
- The most common use of web servers is to host websites, but there are other uses such as data storage or running enterprise applications.
- The primary function of a web server is to deliver web pages on the request to clients using the Hypertext Transfer Protocol (HTTP).

Web Servers - 2

- Some of the major web servers are:
  - Apache
  - Nginx
    (*pronounced,* "**engine-x**")
  - Apache Tomcat
  - IIS
  - Jigsaw
  - Lighttpd
    (*pronounced,* "**lighty**")
  - Klone.
  - Zeus.

Next Action…

The Web Hosting Process

Group Activity – 10 Minutes

- Your group is expected to find out and report on what you think the web design process should be made of.
- Each group represented by one person, shall make a brief presentation on the group's findings.

Good Luck

The Web Design Process

- Designing a web site goes through a series of steps refer to as the Web design process.
- The exact order in which the process is carried varies from organization to organization, and developer to developer.
- Check this site out for details on the web design process.
- http://www.printmag.com/featured/phases-of-the-web-design-process/

Phases of the Web Design Process

- The phases of the web design process include the following steps.

1. **Project Definition**
2. **Project Scope**
3. **Site Architecture**
4. **Visual Design**
5. **Site Development**
6. **Site Testing**
7. **Launching**
8. **Site Maintenance**

Project Definition - 1

- Organizations have a need to communicate to stakeholders their positions on issues and make audiences aware of their products and services.
- Many times, the communication need, such as a web site, is triggered by a change of strategic direction or a new offering.
- Identifying the reasons of the site's existence and what it is supposed to achieve are the first step in the process.
- The goals and objectives that are established at the outset of the project inform all future decisions, from site structure and naming conventions used in the navigation to the visual design of the site.

Project Definition - 2

- The first step in the definition process is interviewing the organization's stakeholders to identify the strategic goals of the site, understand key audience needs and identify key competitors.
- The goal of the definition step is to identify three measurable key outcomes that are directly related to the strategic goals of the organization.
- The challenge in this step is limiting the number of goals.
- Most organizations will have more goals than they know what to do with, and each department believes their individual unit's goals are the most important.
- Being able to bring focus to organizational goals will make developing the site easier and make the final product more effective.

Project Definition - 3

- Once all the information and assessments gathered from the stakeholder interviews are completed, they should be collected in a well-formatted project brief.
- The brief contains the following elements.

1. **Project summary:** Outlines the general overview of the project, organizational background, the environment the organization exists in, the people the organization serves and the unique value it provides to its audience.
2. **Goals:** What are two or three specific measurable goals that the site should achieve? Clear goals allow the Web team the ability to focus on what will provide the most impact and move the organization forward.

Project Definition - 3

3. **Target audiences:** Who will help the organization achieve its stated goals? Most organizations speak to customers, stakeholders, internal audience, suppliers, partners, or government institutions.
4. **Messages:** What are the key messages that attract and motivate key audiences to engage with the organization? What are the key brand messages that help differentiate the organization from its peers?
5. **Competition:** Who are rival organizations that provide similar offerings to your audience? Include an overview of competitive organizations' web sites, considering visual branding, messaging, navigation, calls to action and key differentiators.

Project Scope - 1

- Defining the scope of the project is a critical step.
- One of the most common frustrations with web, and other information system projects is scope creep.
- By creating a well-defined project scope plan that outlines specific activities and deliverables, along with specific timelines, you will be able to clearly set expectations for your clients.
- One of the most common ways of tracking Web projects is through the use of a Gantt chart.

Project Scope - 2

- A Gantt chart not only outlines major activities but also the tasks associated with each activity and start and end dates.
- The Gantt chart provides a visual reference for the team, showing the timeframe of each step and the dependencies between steps.
- The Gantt chart also creates accountability between the Web team and the client (which could be an outside client or simply your boss), letting the client and the team know that the delivery schedule is dependent on everyone hitting their marks; if someone misses a date by a day, the entire schedule shifts by a day.

Site Architecture

- Site architecture includes the sitemap and wireframes of pages.
- Creating the sitemap ensures that you've considered all the key pages in the site, showing their relationship to each other and defining how the sties overall navigation should be structured.
- Wireframes provide a detailed view of the content that will appear on each page.
- Although they do not show any actual design elements, the wireframes provide a guide for defining content hierarchy on the page.

Site Structure - 1

- A good web design requires a solid site architecture based on the site's goals and target audience established in the project brief.
- The deliverables from this phase are:
  - **Content Outline**
  - **Site Diagram**
  - **Wireframes/ Page Description Diagrams**
- These three deliverables need to be completed sequentially.

Site Structure – Content Outline

- Working closely with your clients, create a list of all existing content.
- Brainstorm content that needs to be created for the site.
- Review the list of content, trimming anything that does not match the goals or audience needs as stated in the project brief.

Site Structure – Content Outline

- Take time to think about the future and how the site content might need to grow.
- Make sure you leave room for growth.
- Next group your content into categories.
- Once your categories are established, create an outline of your content and review it with your clients for accuracy.

Site Structure – Site Diagram

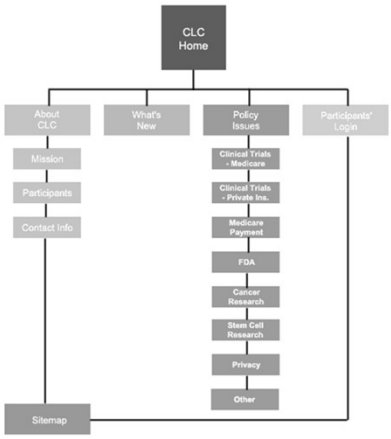
- Take your final content outline and create a sitemap or site diagram.
- A site diagram is just a visual representation of your content outline and site structure.
- You can use MS Word, Visio or other relevant tools to create your site diagram.

Site Structure - Wireframe

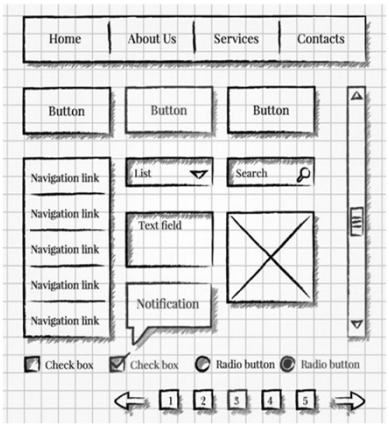
- A wireframe is usually a non-graphical layout of a web page.
- It is a simple drawing of the chunks of information and functionality for each page in your site. You will want to create a wireframe for the home page, each unique second level page and any other significantly different page on the site.
- Wireframes include the containers for all the major elements of the page.
- Elements include navigation, images, content, functional elements (like search) and footer.

Site Architecture

- Site architecture includes the sitemap and wireframes of pages.



Site Map



Wireframe

Visual Design

- Once the blueprint for the site has been defined through the creation of the sitemap and wireframes, the next step is to create a visual style.
- The overall visual style will most likely be determined by the visual brand of the organization; the goal being to connect the Web with all other forms of the organization's communications.
- The organization's brand plays an important role in this part of the process, as designers will want to visually convey key brand perceptual ideas within the design.

Site Development

- With designs approved, it's time to flesh out the design of the pages, develop new content and refine old content, create videos, slideshows, podcasts and other media that will appear on the site as well as start to build out the HTML and CSS of the site.

Site Testing

- Before the site is launched, it will be placed on a production server where only internal audiences and anyone who you share the link with can view it.
- Testing of the site is critical as there will inevitably be issues that need to be addressed before the site goes live.
- There is nothing that erodes a brand more than a site that doesn't function properly or that has misspellings or broken design elements.
- At this stage the site will need to be reviewed on multiple browsers (Firefox, Safari, Internet Explorer) and multiple devices (laptops, tablets, and mobile) to see if and where breaks occur.

Launching the Site - 1

- Once the site is reviewed and approved by the project stakeholders, then you are ready to launch and make it known to the world.
  - ***Register a domain***
  - ***Find a hosting company***
  - ***Upload the website***

Launching the Site - 2

- But once the site is launched, the project isn't over.
- Be prepared to address feedback from users adapting to the new site.
- Expect to make some immediate changes to the site, such as fixing broken links, editing content and making adjustments.
- Note that the Web is a fluid medium that changes on a daily, if not hourly basis — change is inevitable.

Site Maintenance

- Just like any other information system, websites constant care and maintenance.
- Updating content, making changes to the backend and fixing broken links are all in a day's work.
- All of these phases are critical to the Web design process.
- A proper maintenance strategy must therefore be put in place for your website projects.
- This will help organizations achieve a goal, move forward, and prosper in a competitive environment.

Focus Questions?

- What are the key functions of web browsers and web servers.
- In your own perspective, describe what you think should constitute the process of developing a website.
- Outline the steps you would take in hosting your website with a hosting company
- Assume you want to setup your own web hosting company, outline key skillsets, and resources you may need

Our Next Lecture …

1. Overview of HTML
2. HTML Elements and Tags
3. HTML Document Structure
4. Our First Web Page in HTML